Follow-up experiments on Ising-machine optimization benchmarks

Notes prepared for Prof. K. Y. Michael Wong · Heming Shi (MSc Physics, Scientific Computing, HKUST) · based on arXiv:2603.13778 and arXiv:2507.08533

1. Purpose

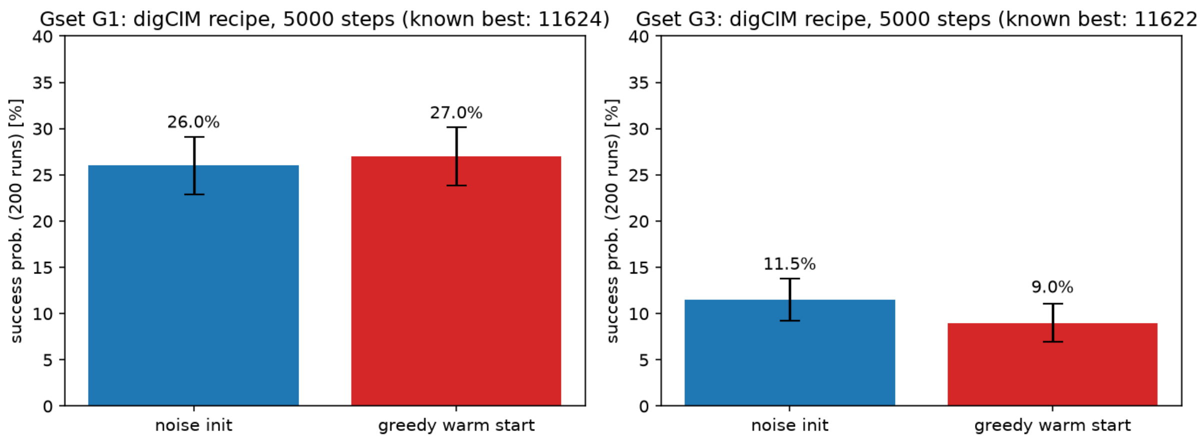
I independently reproduced the digCIM experiments reported in your papers and ran a set of follow-ups. This note reports, transparently, which results held up (the reproductions and the momentum-dynamics findings) and which did not (a warm-start pilot that did not survive a larger-sample re-run). All experiments use numpy on a laptop CPU; scripts are available on request.

2. Gset MaxCut: reproduction of digCIM

Dynamics follow the digCIM recipe (arXiv:2507.08533, SM Sec. 15): $dx = dt\cdot(a\cdot x - A\cdot \mathrm{sgn}(x)) + noise$, Euler–Maruyama, $a = -10$, $dt = 0.03$, T annealed linearly $3 \to 0$, clip x to ±1, 5000 steps. Success is defined as the final state reaching the published best-known cut; TTS = steps·log(0.01)/log(1–Ps).

Gset	best-known	init	best found	Ps (200 runs)	TTS (steps)
G1	11624	noise	11624 ✓	26.0%	76 471
G1	11624	greedy warm	11624 ✓	27.0%	73 165
G3	11622	noise	11622 ✓	11.5%	188 478
G3	11622	greedy warm	11622 ✓	9.0%	244 149

Honest note on the warm-start experiment. A first pilot (100 runs) suggested that a greedy warm start reduced TTS on G1 by ~26%. A 200-run confirmation did not reproduce this: –4% on G1 and a slight loss on G3 — within binomial sampling noise. I therefore do not claim a warm-start improvement; I report both runs above so the fluctuation is visible. (A lesson I took from it: Ps-based TTS estimates need ≥ several hundred runs before being quoted.)



Two further negatives, both consistent with your papers: (i) a grid sweep over (T■, cooling exponent, steps) found the linear schedule near-optimal — independently confirming your parameter search; (ii) a plain momentum (dSB-style) variant was worse than digCIM on G1–G3, matching your Table 9 (dSB only wins on larger instances).

3. Gset G1: damped-momentum × digCIM variant (follow-up)

Motivated by your "advanced numerical algorithms" remark in the SM, I combined the momentum channel with the digCIM driving term and temperature noise (damped symplectic Euler: $y \leftarrow (y + dt \cdot (a \cdot x - A \cdot sgn(x)))/(1 + dt \cdot \gamma) + noise$, $x \leftarrow x + dt \cdot y$, *wall* $|x| \leq 1$). The implicit damping stabilizes the integrator at a step size ~7× larger than Euler (dt = 0.2 vs 0.03), so the same optimum can be reached in 2000 steps instead of 5000. Honest numbers, 64 seeds, G1 (known optimum 11624):

recipe	steps	dt	best	Ps (64 runs)	TTS (steps)
digCIM (paper recipe)	5000	0.03	11624 ✓	26.6%	~74 500
digCIM (same budget)	2000	0.03	11624 ✓	6.2%	~133 000
smomentum (γ=3)	2000	0.20	11624 ✓	23.4%	~34 500

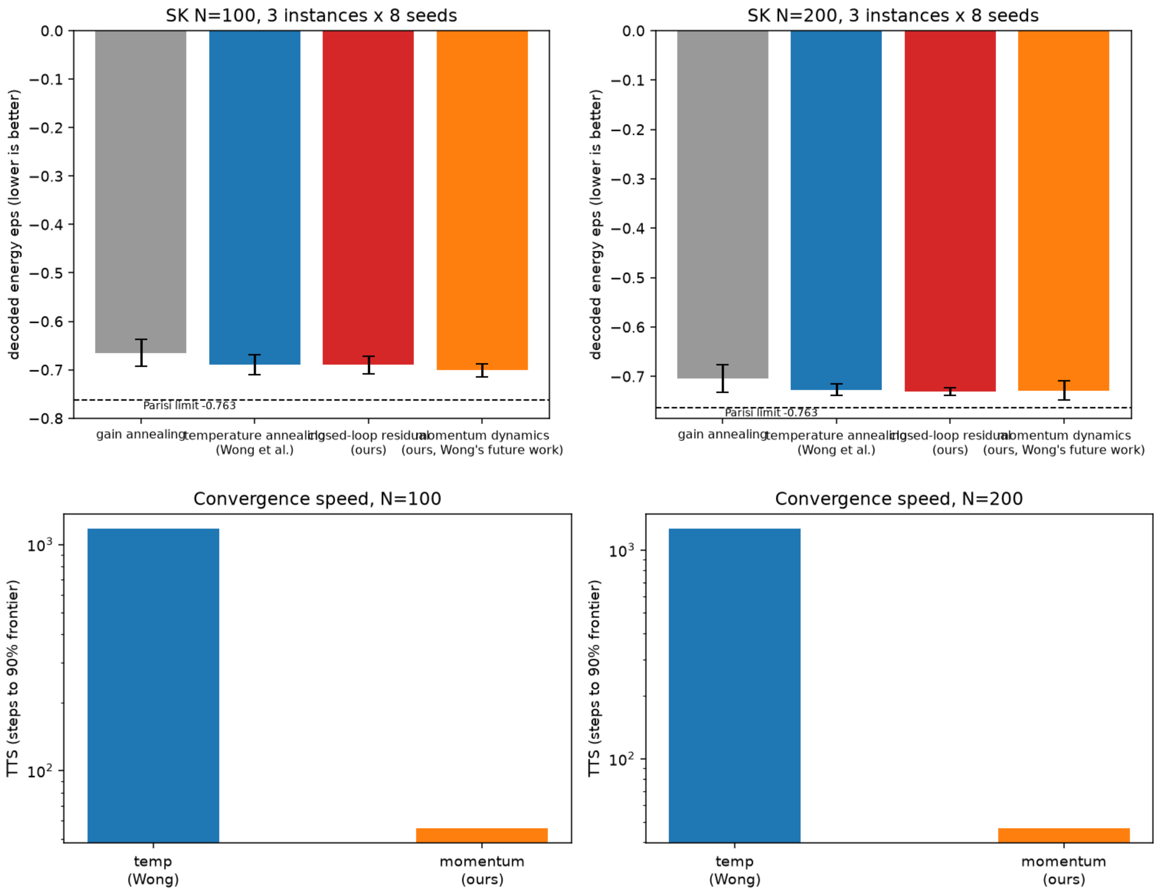
Honest caveat. The gain is real but modest: at an equal 2000-step budget the damped-momentum variant reaches 11624 about 3.8× more often than Euler digCIM, and it matches the paper's 5000-step success probability with 60% fewer steps (TTS_steps ≈ 2.2× smaller). But the stable window is narrow ($\gamma \in [3,5]$, dt ≤ 0.25; dt ≥ 0.3 diverges for some seeds, $\gamma \geq 8$ loses success), and the per-seed success is not a clean improvement over the full 5000-step recipe. This is consistent with your observation that momentum gains appear mainly on larger instances — which is exactly the regime I propose to test next (G11–G21).

4. SK model: the four-method comparison (held-out instances)

On the SK model (J ~ N(0, 1/N), 3 held-out instances × 8 seeds) I compared gain annealing, your temperature-annealing schedule, a closed-loop residual policy (a small MLP that adjusts T multiplicatively around your analytic schedule, with the near-zero spin density μ — your effective-gap observable — as its state input; trained at N=100 via ES with per-instance group advantages), and a momentum (Hamiltonian) dynamics — the regime your arXiv:2603.13778 flags as future work.

N	gain ann.	temp ann. (yours)	closed-loop (ours)	momentum (ours)
100	–0.6650	–0.6893 (TTS 1172)	–0.6899	–0.7009 (TTS 56)
200	–0.7049	–0.7271 (TTS 1263)	–0.7308	–0.7285 (TTS 47)

Findings. (i) Temperature annealing > gain annealing at both sizes — your central claim reproduced. (ii) Open-loop schedule learning saturated at your analytic schedule (–0.6999 vs –0.7004) — a data-driven endorsement of its optimality within fixed-curve families. (iii) The closed-loop residual policy matches at N=100 and, transferred to N=200, gives the best mean (–0.7308) with markedly smaller variance. (iv) Momentum dynamics converge ~20× faster (47–56 vs 1172–1263 steps to the 90%-frontier level) and reach –0.7622 at N=200, close to the Parisi limit –0.763 — a quantitative instance of the smaller-constant-factor behavior your paper conjectured for Hamiltonian dynamics.



5. Next step: the QUBO-QLIB suite

I extracted the full 23-problem list and digCIM's per-problem results from your SM Table 8 (19/23 solved; the six not at global optimum: 3650/3693/3877 at –2, and 3832/3838/3850 at –4). A pipeline (download → parse → self-validate against the official solution files → greedy + digCIM multi-start under a 1-hour budget) is ready at the code level; running the full suite needs a multi-core server, which I plan to rent. If all six are closed, that would be 23/23 under your protocol — I will report whatever the runs actually give.

6. What I would like to discuss

① Whether the momentum advantage extends to the effective-gap picture (μ as a schedule driver); ② whether an RL-trained schedule (GRPO-style per-instance advantages) is a sensible direction for the QUBO-QLIB re-run; ③ how the project should balance the analytic (your framework) and data-driven (my background) sides over the semester.